



Faculty of Computers and Artificial Intelligence
Cairo University

IS211: Introduction to Database

Program: General

Assignment #1 - (third Phase)

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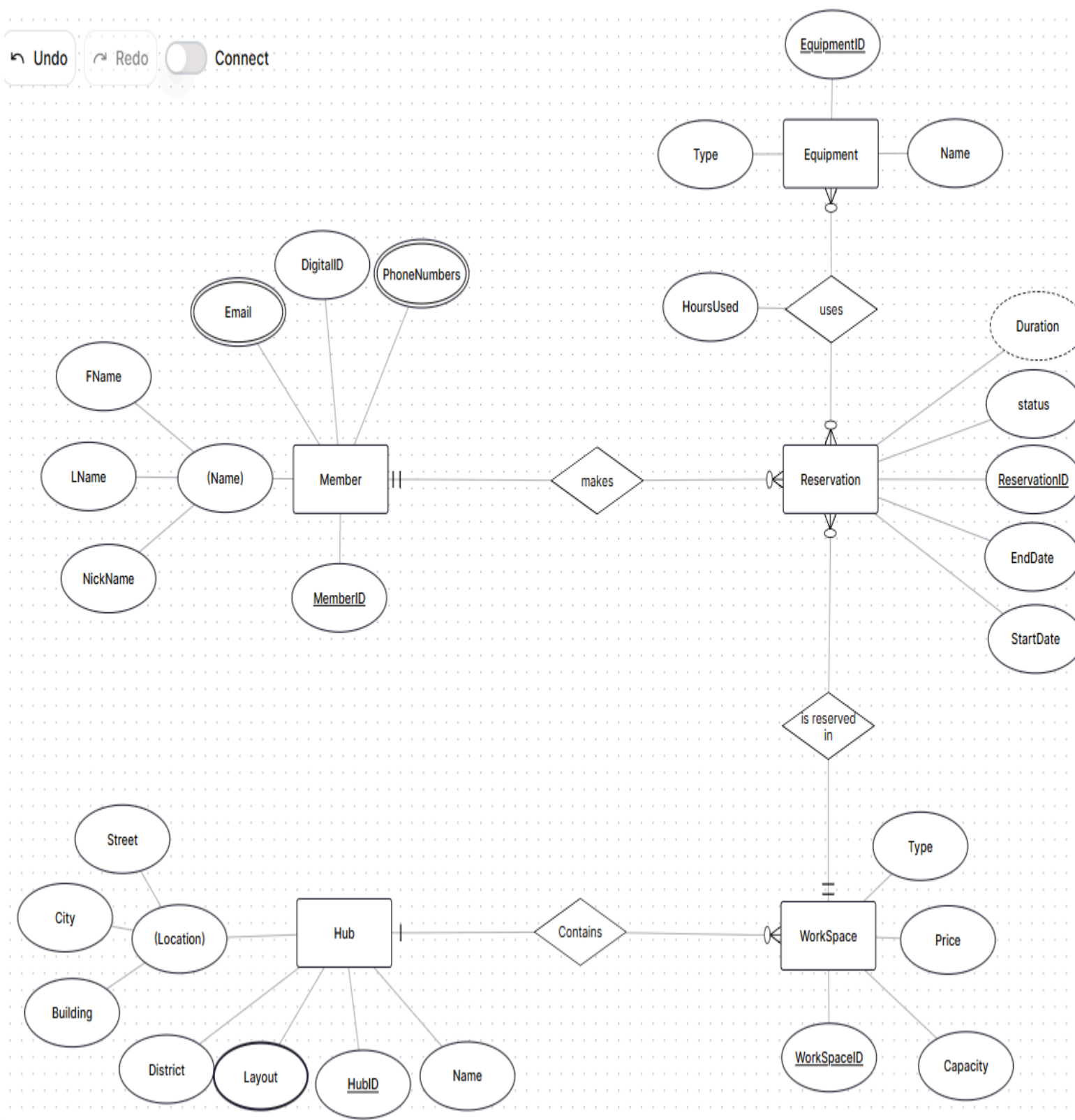
*Project Title: **The Smart Corporate Workspace & Resource Hub***

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ERD-Diagram:



Requirements Specification Smart Corporate Workspace & Resource Hub

1. Introduction

This document defines the system requirements for the Smart Corporate Workspace & Resource Hub. The system is designed to manage coworking environments, including members, workspaces, reservations, and equipment usage across multiple urban hubs.

2. Functional Requirements

2.1 Member Management

- *The system shall allow users to register as members.*
 - *The system shall store member information including Name, Email, Company, and Digital ID.*
 - *The system shall allow updating and managing member profiles.*
-

2.2 Hub Management

- *The system shall store multiple urban hubs.*
 - *The system shall record hub details including Name, Location, District, and Layout.*
-

2.3 Workspace Management

- The system shall manage different types of workspaces (e.g., private offices, open desks, meeting pods).
 - The system shall store workspace details including Type, Price, and Capacity.
 - The system shall associate each workspace with a specific hub.
 - The system shall allow updating workspace availability and pricing.
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2.4 Reservation Management

- The system shall allow members to create reservations for workspaces.
 - The system shall record reservation details including Start Date, End Date, and Status.
 - The system shall allow updating and managing reservation status.
 - The system shall ensure that each reservation is linked to exactly one member and one workspace.
-

2.5 Equipment Management

- The system shall store equipment details including Name and Type.
 - The system shall allow assigning equipment to reservations.
 - The system shall track equipment usage duration for each reservation.
-

2.6 Search and Filtering

- The system shall allow filtering available workspaces by hub location.
 - The system shall allow filtering workspaces by type.
-

3. Data Requirements

- *Each **Member** must have a unique MemberID.*
 - *Each **Reservation** must have a unique ReservationID.*
 - *Each **Workspace** must have a unique WorkspaceID.*
 - *Each **Hub** must have a unique HubID.*
 - *Each **Equipment** must have a unique EquipmentID.*
 - *Each **Reservation** must be associated with exactly one **Member**.*
 - *Each **Reservation** must be associated with exactly one **Workspace**.*
 - *Each **Workspace** must belong to exactly one **Hub**.*
 - *A **Member** may have zero or many **Reservations**.*
 - *A **Workspace** may have zero or many **Reservations**.*
 - *A **Hub** may contain multiple **Workspaces**.*
 - *A **Reservation** may include multiple **Equipment** items.*
 - ***Equipment** may be used in multiple **Reservations**.*
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4. Query Requirements

The system shall support the following queries:

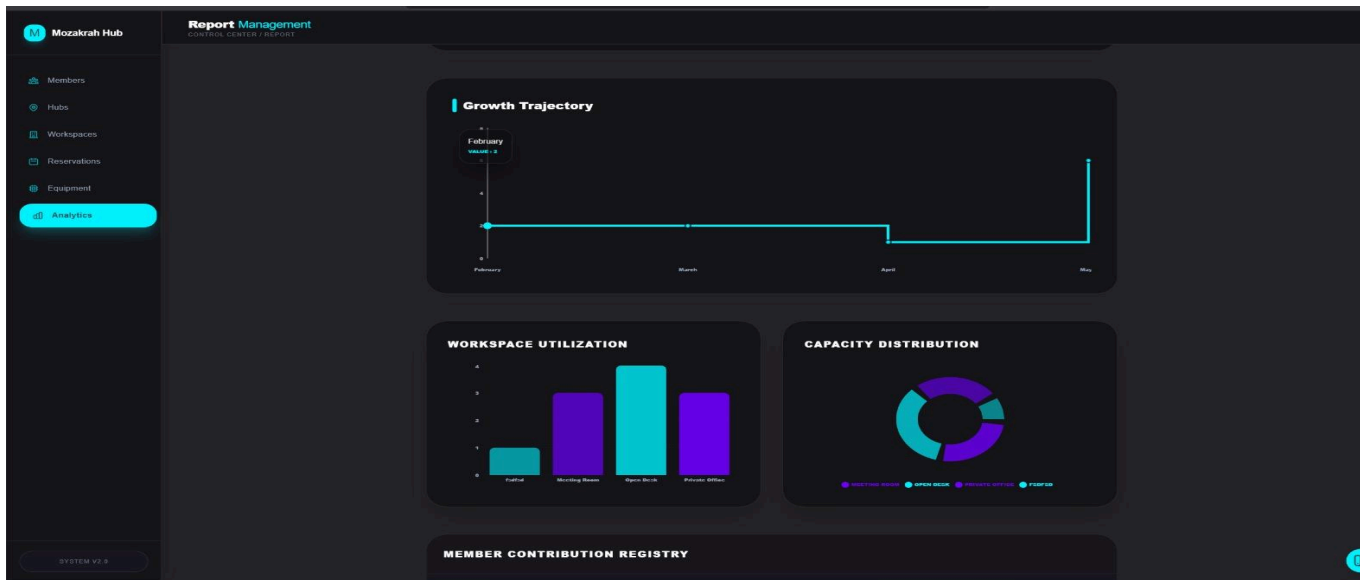
- 1. Determine the most popular workspace type based on the number of reservations.*
 - 2. Identify urban hubs that had no workspace reservations during the last month.*
 - 3. Identify the member who reserved the highest variety of different equipment items during the last month.*
 - 4. Identify members who did not make any workspace reservations during the last month.*
 - 5. List all equipment items utilized in each urban hub during the last month.*
 - 6. Retrieve each member's full profile along with the total number of hours they reserved.*
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5. Assumptions

- *Reservation duration is calculated as the difference between Start Date and End Date.*
 - *Equipment usage duration is recorded separately for accurate tracking and billing.*
 - *All members must be registered before making reservations.*
 - *Each workspace is associated with exactly one hub.*
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6. Conclusion

This document outlines the essential functional, data, and query requirements for the Smart Corporate Workspace & Resource Hub system. These requirements serve as the foundation for designing the system's database and ERD.





Hubs Management

CONTROL CENTER / HUBS



Establish New Hub

HUB NAME

e.g. Neo Tokyo

CITY

Metropolis

DISTRICT

Central Sector

STREET

Cyber Avenue 101

BUILDING

Unit 7B

LAYOUT INFO

